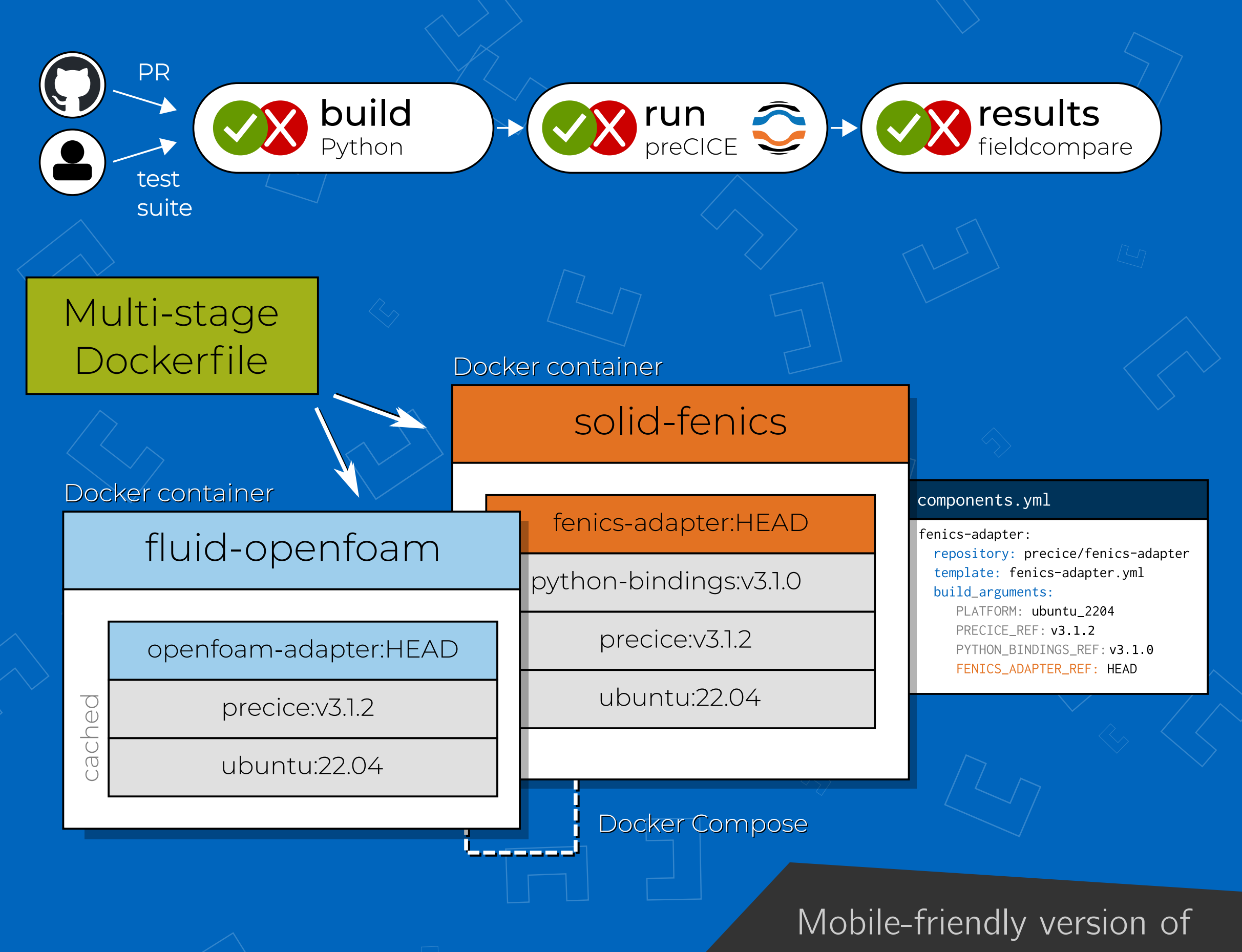
Ensuring your simulations still run correctly



components.yml

```
fenics-adapter:
  repository: precice/fenics-adapter
  template: fenics-adapter.yml
  build_arguments:
    PLATFORM: ubuntu_2204
    PRECICE_REF: v3.1.2
    PYTHON_BINDINGS_REF: v3.1.0
    FENICS_ADAPTER_REF: HEAD
```

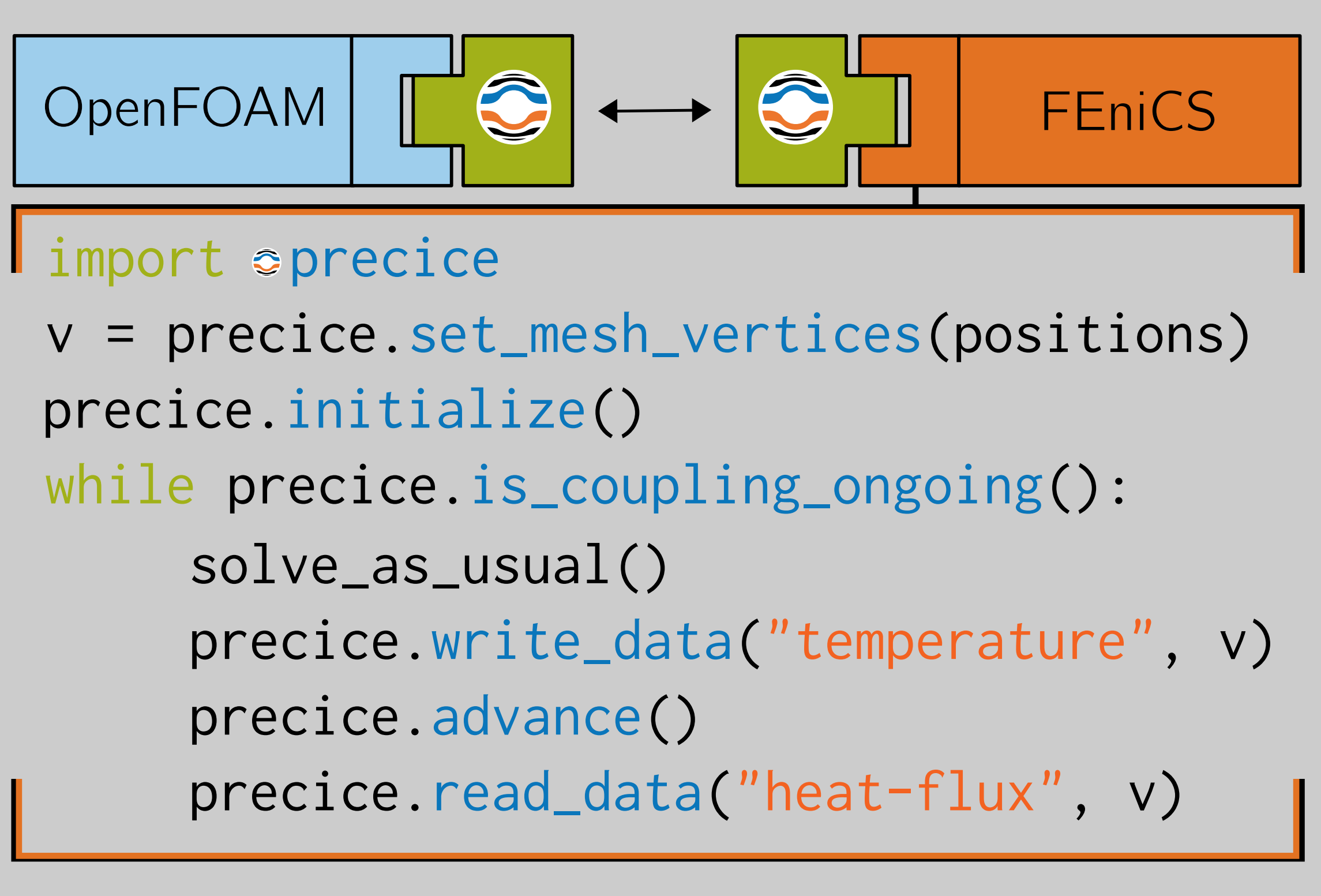
Mobile-friendly version of poster presented at the preCICE Workshop 2024 September 24-27, Stuttgart

Plain language summary

The software library preCICE allows programmers and end users to combine simulation models with minimal effort. Integrations with simulation codes are developed by multiple people in various places, and useful simulations can only be executed by combining multiple of theses codes. To test that development changes do not affect the numerical results, we create reproducible environments from a common description and for the versions of interest, connect them, and check the results at the interface between the individual simulations. With this mechanism in place, the community can now shape their contributions in a way that can be continuously tested and easily maintained.

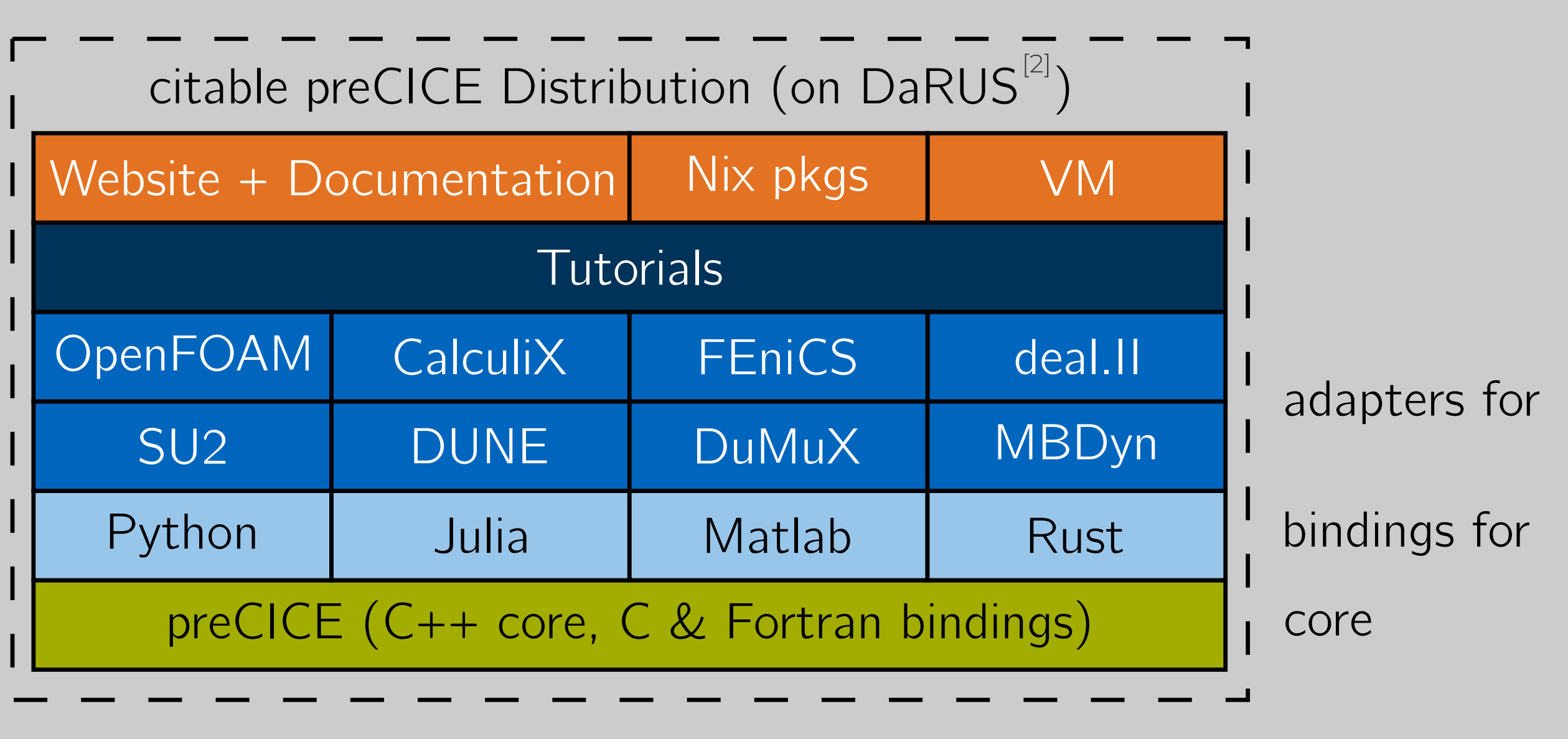
Coupling via preCICE

Add calls to the preCICE library into each code, or use provided adapters. Start each code in a separate process (e.g., two terminals).



```
import precice
v = precice.set_mesh_vertices(positions)
precice.initialize()
while precice.is_coupling_ongoing():
      solve_as_usual()
      precice.write_data("temperature", v)
      precice.advance()
      precice.read_data("heat-flux", v)
```

A citable ecosystem of components



citable preCICE Distribution (on DaRUS[2])

Website + Documentation		Nix pkgs	VM	
Tutorials				
OpenFOAM	CalculiX	FEniCS	deal.II	adapters for
SU2	DUNE	DuMuX	MBDyn	
Python	Julia	Matlab	Rust	bindings for
preCICE (C++ core, C & Fortran bindings)				core

Community contributions

preCICE store
O
C

Developing guidelines

☐ Metadata, citing
☐ Tests available
☐ Reference results
☐ Executing via scripts

preCICE conforming Gold level
preCICE conforming Silver level
preCICE conforming Bronze level

Reference results and comparisons

Stored next to the cases

Git Large File Storage

git-lfs.com

VTK compared with

FieldCompare

gitlab.com/dglaeser/fieldcompare

```
flow-over-heated-plate/                                # One directory for each example
    fluid-openfoam/                                    # One directory for each code
    solid-fenics/
    solid-nutils/                                      # Multiple code options
    reference-results/                                 # Reference results versioned
        fluid-openfoam_solid-fenics.tar.gz             # One file for each combination
        fluid-openfoam_solid-nutils.tar.gz
```

fluid-openfoam	preCICE	solid-fenics
OpenFOAM output .foam, logs	mesh exports (.vtk) iterations.log	FEniCS output .vtk, logs
specific to OpenFOAM, controlled by others	standard formats (VTK, CSV)	specific to FEniCS, controlled by others

Metadata, tests, and test suites

metadata.yaml (precice/tutorials/flow-over-heated-plate/)

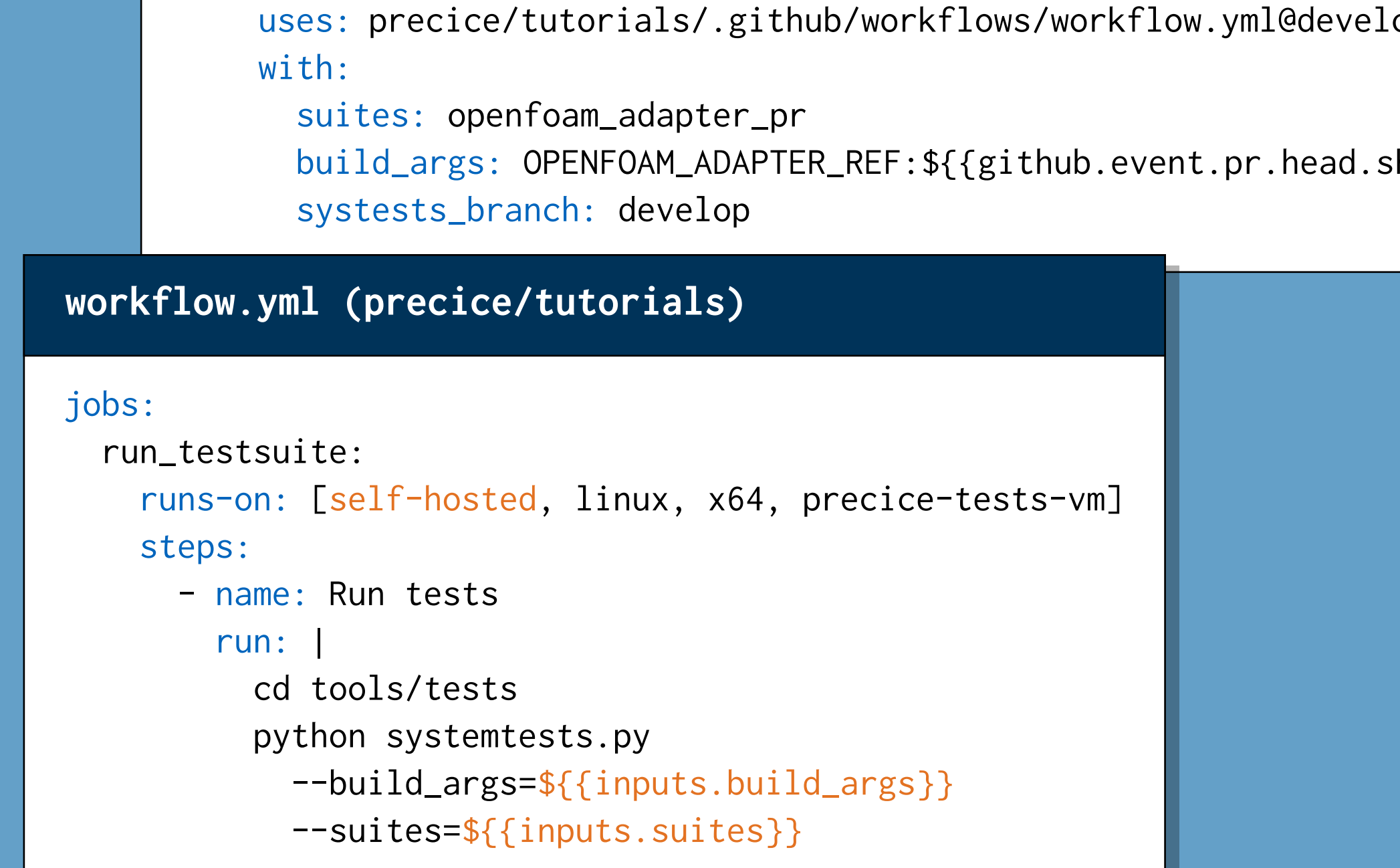
```
name: Flow over heated plate
path: flow-over-heated-plate

participants:
  - Fluid
  - Solid

cases:
    fluid-openfoam:
      participant: Fluid
      directory: ./fluid-openfoam
      run: ./run.sh
      component: openfoam-adapter

    solid-fenics:
      participant: Solid
      directory: ./solid-fenics
      run: ./run.sh
      component: fenics-adapter
```

tests.yaml (precice/tutorials)

```
test_suites:
  openfoam_adapter_pr:
    tutorials:
      - path: flow-over-heated-plate
        case_combination:
          - fluid-openfoam
          - solid-fenics
        reference_result:
          fluid-of_solid-fenics.tar.gz
```

Multiple repos, multiple perspectives

Partitioned approach makes CI complex. Tests can run locally or on GitHub Actions, triggered manually or by pull requests.



	nightly	on PR	
preCICE	develop	v3.1.2	
Python	develop	v3.1.0	
FEniCS	develop	v2.1.0	
OpenFOAM	develop	v1.3.1	feature
Tutorials	develop	v202404.0	

Can we publish a new release? Can we merge the PR?

openfoam-adapter-workflow.yml (precice/openfoam-adapter)

```
jobs:
  gather-refs: (map from branch to latest commit hash)
  run-system-tests:
    if: ${{github.event.label.name == 'trigger-system-tests' }}
    uses: precice/tutorials/.github/workflows/workflow.yml@develop
    with:
      suites: openfoam_adapter_pr
      build_args: OPENFOAM_ADAPTER_REF:${{github.event.pr.head.sha}}
      systests_branch: develop
```

workflow.yml (precice/tutorials)

```
jobs:
  run_testsuite:
    runs-on: [self-hosted, linux, x64, precice-tests-vm]
    steps:
      - name: Run tests
        run: |
          cd tools/tests
          python systemtests.py
            --build_args=${{inputs.build_args}}
            --suites=${{inputs.suites}}
```

References

[1] **preCICE v2**
G. Chourdakis, K. Davis, B. Rodenberg, M. Schulte, F. Simonis, B. Uekermann, et al. (2022), preCICE v2: A sustainable and user-friendly coupling library [version 2; peer review: 2 approved] Open Res Europe, 2:51, https://doi.org/10.12688/openreseurope.14445.2

[2] **preCICE Distribution**
J, Chen, G. Chourdakis, I. Desai, C. Homs-Pons, B. Rodenberg, D. Schneider, F. Simonis, B. Uekermann, K. Davis, A. Jaust, M. Kelm, N. Kotarsky, H. Kschidock, D. Mishra, M. Mühlhäußer, T. P. Schrader, M. Schulte, V. Seitz, J. Signorelli, G. van Zwieten, N. Vinnitschenko, T. Vladimirova, L. Willeke, E. Zonta
(2024) preCICE Distribution Version v2404.0. https://doi.org/10.18419/darus-4167, DaRUS, V1.

[3] **preCICE System Tests**
G. Chourdakis, V. Seitz, B. Uekermann (2024)
System regression tests for the preCICE coupling ecosystem
Proceedings of the deRSE24, Electronic Communications of the EASST (in review)

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The preCICE system regression tests

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